Properties of Individual Dopant Atoms in Single-Layer MoS_2: Atomic Structure, Migration, and Enhanced Reactivity

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Dopants generally govern physical and chemical properties of all kinds of solids, especially semiconductors where doping is widely used to adjust carrier densities and tailor the electronic characteristics of the devices. The effects of doping become more significant in case of two-dimensional (2D) materials, as in addition to the substitutional doping, carrier concentration can be controlled by adding adatoms or layers of another material. For the case of graphene, the Fermi level can be tuned efficiently by metal contact[1] or noncovalent chemical functionalization.[2] Recently, layered inorganic transition-metal dichalcogenides (TMDs) exhibiting a wide variety of electronic properties[3–5] have received lots of attention. Among them, single-layer MoS_2 possesses a direct gap[6,7] and exhibits carrier mobility comparable to the graphene nano-ribbons with high current on-off ratios.[8] The electrical conductivity of MoS_2 may then be further modulated by substitutional doping, such as Re (n-type)[9–11] and Nb (p-type).[12,13] The MoS_2 sheets with adsorbed Au particles show a remarkable p-type character.[14,15] Besides, intercalation of noble metals (Au or Ag) in MoS_2 nanotubes has been reported[16,17] so that the knowledge of how dopant atoms interact with the host TMD crystal lattices is highly important in the context of both fundamental science and future device technology.

The Z-contrast annular dark-field (ADF) imaging by means of STEM is a powerful method to identify not only the structure but also chemical composition atom-by-atom. Voyles[18] visualized individual Sb atoms and clusters in bulk Si, and our previous work reported a random distribution of hetero-transition metals in the single-layer $Mo_{1-x}W_xS_2$ alloy.[19] Recently, the observations of individual B and N dopants as well as single three or four coordinated Si impurity atoms in graphene lattice were reported.[20–22] Besides these, the dynamics of single Fe atom and Si cluster in graphene vacancies under the electron beam were also monitored in TEM.[23,24] Finally, the presence of Co and Ni dopants at the edges of MoS_2 hydrotreating catalysts was demonstrated in scanning tunneling microscopy studies.[25,26] However, no direct experimental information on the atomic configurations of dopants in 2D TMDs is available so far. Moreover, correlations between their dynamic behavior and nano-scale properties have never been studied before.

Here we perform an STEM study focused on the atomic structures of single-layered MoS_2 doped with Re and Au. By combining the experimental STEM observations with first-principles calculations carried out within the framework of the density functional theory (DFT), we unequivocally identify the atomic positions of Re and Au dopants in the host MoS_2 lattice and correlate the migration behaviors of dopant atoms with their energetics. Dynamic observations of the atomic structure evolution reveal the interaction of the dopant atoms with impurity atoms deposited on the doped MoS_2 sheet, which proves local enhancement of the chemical affinity of the system.

Typical ADF images of single-layered Re-doped and Au-doped MoS_2 are presented in **Figure 1**a and 1b, respectively. The dopants, Re (Z = 75) and Au (Z = 79), appear in brighter contrast in the ADF images than both Mo (Z = 42) and S (Z = 16) as indicated by arrows. Chemical analysis by means of EDX was also done to confirm the doping elements (Figure S2 and S3). The high resolution ADF image in the inset of Figure 1a clearly shows that Re atoms sit at the Mo sites. The Re dopants are well dispersed in MoS_2 layers and seldom form clusters on the host material. The observed doping concentration is around ~0.6 at% Re in agreement with the synthesis condition. On the other hand, the Au dopants at similar concentration (~0.6 at% Au) tend to aggregate on the MoS_2 surface (open circles). The Au atoms are mobile under the electron beam and can be found either on top of the Mo, S, or the hollow center (HC) of MoS_2 hexagons, as shown in the inset of Figure 1b.

In order to understand the dynamic behaviors of Re dopants in MoS_2, we analyzed ~450 Re dopant atoms. 89% of the Re atoms (~400) are substitutional dopants at the Mo sites (Re@Mo), and they did not migrate at all throughout the acquisition time (typically 240 s). We found that the rest 11% of Re were adatoms (~51) and changed their positions in sequential images. **Figures 2**a–c show a set of sequential ADF images illustrating the migration of four Re dopant atoms in a MoS_2 single-layer. Three Re atoms (Re@Mo) remained stationary

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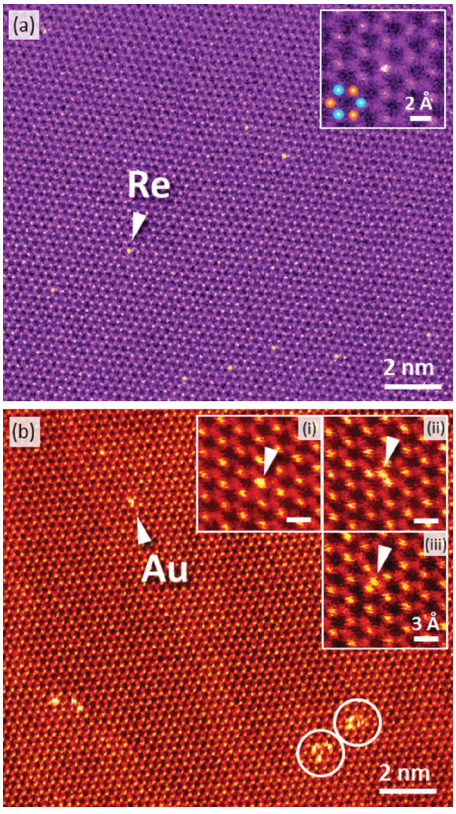


Figure 1. (a) Large area ADF image of clean Re-doped single-layer MoS_2. The inset shows a Re dopant substitution at the Mo site with atomic resolution as evident from a brighter contrast. Blue and orange balls indicate Mo and two overlapped S atoms, respectively. There are 10 Re atoms (one pointed by the arrow) found in ~1500 MoS_2 unit cells, which corresponds to a doping concentration of ~0.6 at%. (b) Large area ADF image of ~0.6 at% Au-doped single-layer MoS_2. The insets (i), (ii), and (iii) show Au adatom (indicated by arrows) located on top of the Mo, S, and HC site, respectively.

for the whole acquisition time (up to t = 335.4 s). In Figure 2b, the Re adatom on S (Re-S, indicated by open arrow) jump to the Mo site (Re-Mo) and then move to another S site (Re-S) (Figure 2c) to form again the complex Re-S+Re@Mo dopant structure (Movie M1). Interestingly, the Re-S structure always appears in the vicinity of Re@Mo, and therefore forms the dopant structure as a complex of Re-S+Re@Mo among 17 cases observed. The experimental ADF image (Figure 2h) was corroborated by STEM simulation with DFT calculated atomic model (Figure 2g) using the appropriate the experimental parameters. The simulated ADF profile shows a perfect match with the experiment (Figure 2i), if possible specimen inclination is taken into account. Other observed Re doped structures are summarized in Figure S4.

On the contrary, 95% of Au dopants are highly mobile adatoms easily migrating under the electron beam, and the rest 5% of Au dopants are stable Au@Mo complexes in the dataset of 220 Au dopant atoms observed. Figures 2d–f show sequential ADF images of Au migrations (Movie M3). One Au-S atom in the single layer region is continuously moving and has a different position in each frame. Another Au atom continues to migrate in the bilayer region. The bilayer MoS_2 region shows unconventional rhombohedral stacking (Figure S7), which probably comes from the intercalation of Au atoms.[27] It was not possible, though, to unambiguously determine if the Au atoms were migrating between the layers.[16,17] Figures 2j-l show the identification of Au-S structure by comparing the ADF profile with the STEM simulation. The other Au dopant structures are summarized in Figure S5 and S6. Statistically, most of the Re atoms are in stable Re@Mo positions, and Au atoms are mobile adatoms (Figure S8).

To confirm the migration behavior of the dopant atoms under electron beam, we carried out spin-polarized DFT calculations of doped MoS_2 layers using the VASP simulation package[28,29] as described in supporting information and computed the energetics of dopant atoms in the subsitutional and adatom positions. The formation energies (E_f) of a dopant atom in adatom and substitutional configurations was calculated as

$$E_f = E^D_{total} - (E_{total} - \delta\mu_Y + \mu_D) \quad (1)$$

where E^D_{total} and E_{total} are the total energies of the doped and pristine supercells; μ_D and μ_Y are the chemical potentials of a single dopant (D = Re or Au) and the substituted atom in the host lattice (Y = Mo or S), respectively. μ_Y and μ_D were chosen to correspond to the bulk phase of each element. In the case of adatom, μ_Y term was omitted: $\delta = 0$; while $\delta = 1$ in Equation (1) corresponds to the case of the substitutional dopant.

The formation energies of dopant atom in different configurations are presented in **Table 1**. The values of E_f give an idea of what the favored sites for the dopants are after high-temperature CVT growth, which is relatively close to thermal equilibrium conditions. In the case of Re dopants, substitution to Mo site has the lowest formation energy by a large margin. Among the adatom sites, the position over Mo is favored. This is in a good agreement with the observations that nearly all Re atoms are located at Mo sites, and rarely on the adatom sites on top of Mo and S. In a stark contrast, substituting Mo with Au is highly unfavorable, and the Au atoms are expected to be located predominantly at the adatom sites, which is also consistent with our experimental findings. We calculated the migration barrier for Au adatom diffusion between the lowest energy S-site and the second lowest energy HC-site which proved to be very close to the formation energy difference ~40 meV. It should be pointed out, though that DFT with semilocal exchange and correlation functionals generally underestimates the barriers,[30] and besides the diffusion may be somewhat suppressed by light atoms present on the surface, which are not visible in the TEM images. Several possible configurations for the complex structures with two Re atoms were considered (Table S1). The complex dopant structure is energetically more stable than the two isolated ones according to the DFT calculations.

Local densities of states (LDOS) around the dopant atoms were also calculated for all the experimentally found atomic configurations including the very rare occasions as shown in **Figure 3**. Re dopants substituting Mo atoms provide extra electrons into the conduction band of the MoS_2 (n-type). On the other hand, Au dopants in/on MoS_2 give rise to similar mid-gap states at all the sites and may compensate n-type doping. For larger Au clusters, the work function is expected to lie close to valence band, subsequently leading to p-type doping.[14]

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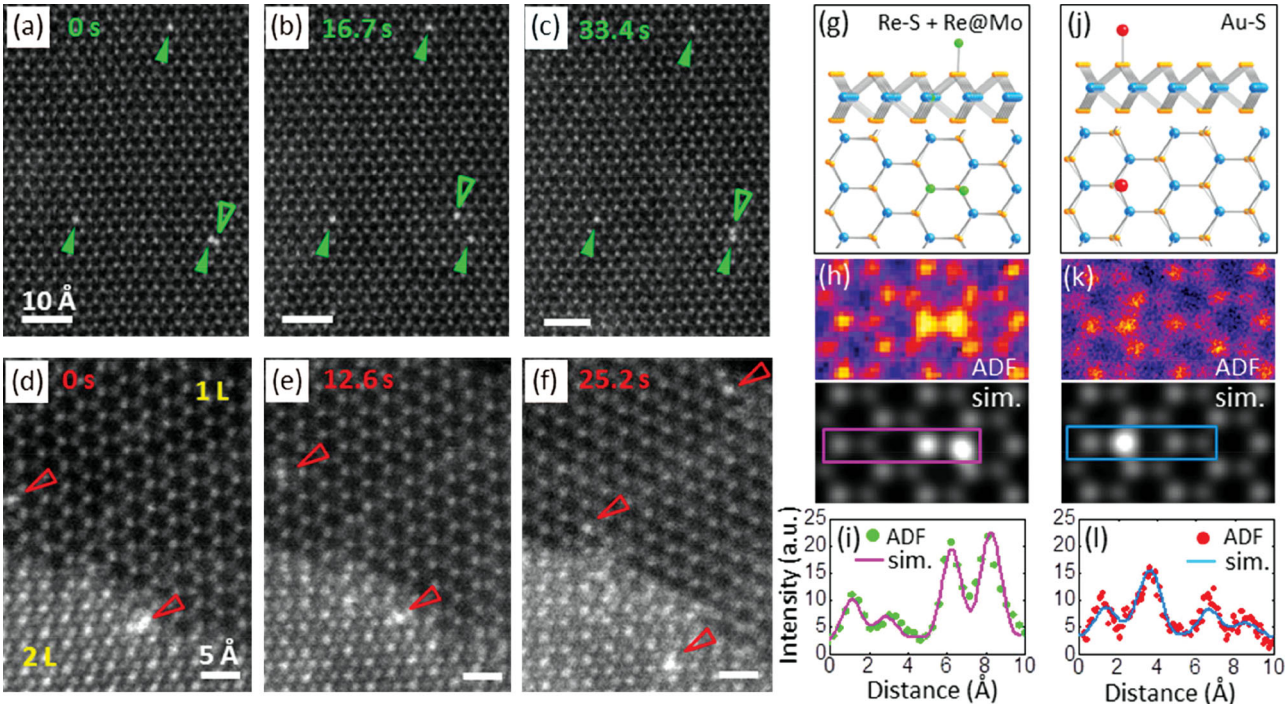


Figure 2. Sequential ADF images of single-layer Re-doped and Au-doped MoS_2. (a) Three Re@Mo are indicated by green arrows and one Re-S adatom is indicated by the open arrow. (b) At t = 16.7 s, the Re-S adatom moves to the Mo site as a Re-Mo and (c) move to another S site as a Re-S again at t = 28.5 s. (d-f) Au adatoms on single- and bilayer regions (indicated by red open arrows) change their atomic positions in every frame. The bilayer region shows rhombohedral stacking [see Fig. S4]. (g) (j) The atomic model of Re-S+Re@Mo complex dopant structure and Au-S. (h) (k) The corresponding ADF images and simulated images of Re-S+Re@Mo and Au-S. (i) (l) The profile of experimental ADF counts and fitting of simulation line profiles for the two cases.

Interestingly, our simulations suggest that Au adatoms can possess magnetic moments on MoS_2 surfaces, so that the introduction of Au impurities may lead to new functionalities of MoS_2 and likely other TMDs. Magnetic moments of Re adatoms were generally larger than for Au adatoms: 3, 3, and 5 μ_B for the Mo-, S-, and HC-sites, but their contribution should be very small due to their rarity in our samples. Moreover, gold nanoparticles on MoS_2 were reported to dramatically enhance the photocurrent.[31]

In order to prove that the dopants indeed alter the local physical/chemical properties of MoS_2 layers, we took advantage of the inevitable carbon deposition in the TEM chamber, while the surface coverage rate of carbon can be reduced after 15 min of e-beam shower. Interesting interaction between carbon contaminations and dopants was recorded afterwards on a reasonable time scale. In **Figure 4**, a sequence of STEM images illustrates the observation of the coverage with amorphous carbon on MoS_2 surface. The dopants were indeed found to behave as defective sites in the MoS_2 lattice, which caused lower adsorption energy and provided anchoring points for accumulating amorphous carbon. The carbon coverage starts in the doped area (Figure 4a to 4c). By *in situ* EELS monitoring (Figure S10), we confirmed that the amorphous carbon moved and eventually anchored at the Re dopant sites. The carbon coverage areas are very localized and extend over a few atomic distances from the dopant sites only. This phenomenon directly proves that the dopants cause the alteration of local physical/chemical properties of MoS_2 layers.

The C agglomeration process is further corroborated by DFT calculations. We first note that the calculated C adatom diffusion

Table 1. Formation energy of Re and Au adatom/substitutional defect complexes on Mo, S, and HC sites, as calculated using DFT. E_k is the maximum kinetic energy of each dopant atom received from 60 kV electron beam.

		Adatom [eV]			Substitutional [eV]	
Dopant	E_k (eV)	Mo	S	HC	Mo	S
Re	0.75	6.09	7.37	7.28	1.9	6.0
Au	0.71	2.46	2.33	2.37	6.9	3.1

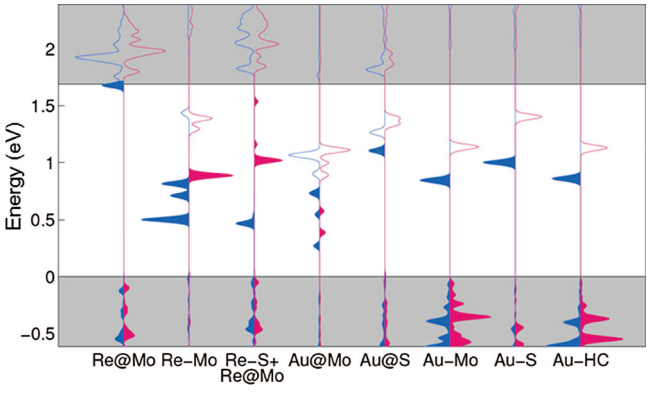


Figure 3. Local density of states of the Re and Au dopant atoms at eight considered atomic structures. The blue and red lines represent the spin-up and spin-down components, and the electron occupied states are shown as filled curves. The grey areas denote the valence band (lower) and conduction band (upper) energy ranges.

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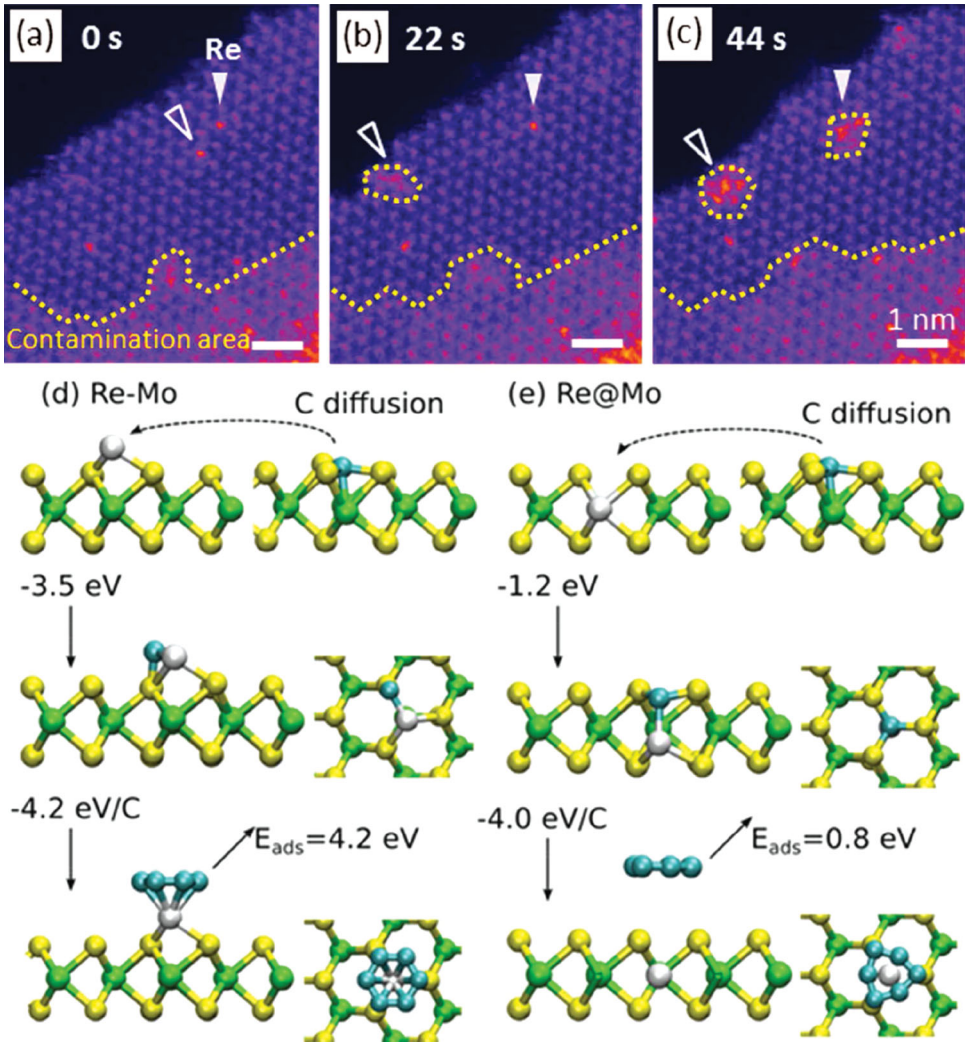


Figure 4. Dynamic behavior of carbon contaminations on Re-doped MoS_2 surface. (a-c) Three sequential ADF images with Re dopants are indicated by arrows. The yellow dotted regions indicate the area covered by amorphous carbon contaminations. (d-e) Illustration of the C segregation processes around Re-Mo (d) and Re@Mo (e) impurities, and the related energies as obtained from DFT calculations. E_{ads} corresponds to the energy required to remove the molecule from the surface.

barrier on MoS_2 is 0.81 eV, evidencing that C atoms are mobile under the experimental conditions. As illustrated in Figures 4d and 4e, the C atom agglomeration around both Re@Mo and Re-Mo is energetically favored. Due to its under-coordination, the Re-Mo adatom binds strongly to the surrounding C atoms even in the case of C_6 hexagon. This is in agreement with experiments, where clusters formed more readily around the Re-Mo impurities. Also in the case of Re@Mo, the C adatoms prefer the sites around the Re impurity, but once the C atoms form larger stable clusters, their binding to the impurity becomes fairly weak and is likely dominated by van der Waals forces for larger flakes. In fact, on the basis of these results, we envisage possibility of Re@Mo behaving as a chemically active site.

In summary, the successful doping of MoS_2 with Re and Au has been demonstrated suggesting the possibility of nano-electronic devices made of doped single-layered MoS_2. In comparison with the graphene electronics, the doped MoS_2 layers will find broader applications with their suitable bandgaps for nano-electronics. We performed in situ experiments providing direct observations of how single dopant atoms interact with other dopants or impurities (such as carbon). Re substitution in Mo site is proved to lower the doping energy for adjacent sites, which results in Re adatom segregation and may explain the difficulties in achieving a high compositional rate of homogeneous Re doping in MoS_2. Because the doping sites exhibit local enhancement of chemical reactivity, they likely provide catalytically active sites. Furthermore, the doped MoS_2 surface can be used as an ideal specimen support to fix individual molecules for molecular imaging or bio-sensor measurements.[32]

Experimental Section

MoS_2 single-crystals doped with 0.5–1 at% of Re or Au (purity: 99.99%) were synthesized by the chemical vapor transport (CVT) method with Br_2 and I_2 as transport agents, respectively.[9,27] The suspended single-layered MoS_2 flakes were then prepared by micromechanical cleavage

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and transferred to a TEM grid (quantifoil) (Figure S1). ADF imaging was carried out in an aberration-corrected STEM JEOL-2100F equipped with a DELTA corrector and a cold field emission gun operated at 60 kV which is theoretically below the threshold of knock-on damage of MoS_2.[33] The scanning rate was set ranging widely from 8 to 128 μs/pixel and beam current 15~20 pA. The dwell time per pixel was 16~58 μs and the pixel size 0.07–0.137 Å for taking movies (sequential images), corresponding to the electron dose of $1.5\sim7.2 \times 10^5$ e/Å^2. The pixel time was increase to 76–128 μs for taking single high resolution images at low magnifications (<10 M). The electron beam damage on MoS_2 lattice would be observed when the total electron dose exceed 5–11 $\times$ 10^6 e/Å^2. We observed a similar beam-damage threshold as in the report from Novoselov and co-workers,[34] but we could achieve longer acquisition times due to the lower beam current used in our experiments. The EDX was performed by a JEOL ARM-200F microscope operating at 80 kV with approximately 0.16 nm probe of 200 pA, 0.5 second/pixels, 50 steps line scan

The single-layered MoS_2 flakes transferred from the SiO_2 surface to the quantifoil by 2-Propanol and cleaned by chloroform. Then the specimens were cleaned by baking in air at 200°C for 10 min, after which they were placed in the TEM chamber. The JEOL double tilt holder was pre-cleaned by ion-plasma (JEOL ion cleaner, 360 V, 10 min) after ethanol treatment. The vacuum level in the TEM chamber was ~1.7×10^{-5} Pa; 15 min of high current e-beam shower (current density ~107 pA/cm^2) was introduced to fix the mobile amorphous carbon and to reduce its migration toward the e-beam scanned area.

Image simulations were carried out by using the MacTempasX software packages. The ADF images were recorded at a convergence angle of α = 24 mrad and an annular detector inner semiangle of β_1 = 55 mrad. STEM simulation parameters were using a probe size of 1 Å, Cs = 1 μm, spread of defocus = 53 Å, and microscope focus value = -2 nm with possible specimen inclinations taken into account. For the DFT calculation, the system was modeled using 5 $\times$ 5 supercell of single-layered MoS_2. The Perdew-Burke-Ernzerhof form of the exchange-correction functional[35] was used. Brillouin zone sampled with a 2 $\times$ 2 $\times$ 1 k-point mesh and 500 eV plane wave cutoff were found to yield converged results. Atom diffusion barriers were calculated using the Nudged Elastic Band method.[36]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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[1] G. Giovannetti, P. A. Khomyakov, G. Brocks, V. M. Karpan, J. van den Brink, P. J. Kelly, *Phys. Rev. Lett.* **2008**, *101*, 026803.
[2] T. O. Wehling, K. S. Novoselov, S. V. Morozov, E. E. Vdoin, M. I. Katsnelson, A. K. Geim, A. I. Lichtenstein, *Nano Lett.* **2008**, *8*, 173.
[3] M. Chhowalla, H. S. Shin, G. Eda, L. J. Li, K. P. Loh, H. Zhang, *Nat. Chem.* **2013**, *5*, 263.
[4] Q. H. Wang, K. Kalantar-Zadeh, A. Kis, J. N. Coleman, M. S. Strano, *Nat. Nanotech.* **2012**, *7*, 699.
[5] W. Zhou, X. Zou, S. Najmaei, Z. Liu, Y. Shi, J , Kong, J. Lou, P. M. Ajayan, B. I. Yakobson, J. C. Idrobo, *Nano Lett.* **2013**, *13*, 2615.
[6] S. W. Han, H. Kwon, S. K. Kim, S. Ryu, W. S. Yun, D. H. Kim, J. H. Hwang, J. S. Kang, J. Baik, H. J. Shin, S. C. Hong, *Phys. Rev. B* **2011**, *84*, 045409.
[7] K. L. Mak, C. Lee, J. Hone, J. Shan, T. F. Heinz, *Phys. Rev. Lett.* **2010**, *105*, 136805.
[8] B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, A. Kis, *Nat. Nanotechnol.* **2011**, *6*, 147.
[9] K. K. Tiong, Y. S. Huang, C. H. Ho, *J. Alloys Compd.* **2001**, *317–318*, 208.
[10] F. L. Deepak, R. Popovitz-Brio, Y. Feldman, H. Cohen, A. Enyashin, G. Seifert, R. Tenne, *Chem. Asian. J.* **2008**, *3*, 1568.
[11] L. Rapoport, A. Moshkovich, V. Perfilyev, A. Laikhtman, I. Lapsker, L. Yadgarov, R. Rosentsveig, R. Tenne, *Tribol. Lett.* **2012**, *45*, 257.
[12] V. V. Ivanovskaya, A. Zobelli, A. Gloter, N. Brun, V. Serin, C. Colliex, *Phys. Rev. B* **2008**, *78*, 134104.
[13] F. L. Deepak, H. Cohen, S. Cohen, Y. Feldman, R. Popovitz-Brio, D. Azulay, O. Millo, R. Tenne, *J. Am. Chem. Soc.* **2007**, *129*, 12549.
[14] Y. Shi, J. K. Huang, L. Jin, Y. T. Hsu, S. F. Yu, L. J. Li, H. Y. Yang, *Sci. Rep.* **2013**, *3*, 1839.
[15] T. S. Sreeprasad, P. Nguyen, N. Kim, V. Berry, *Nano Lett.* **2013**, *13*, 4434.
[16] M. Remskar, Z. Skraba, P. Stadelmann, F. Levy, *Adv. Mater.* **2000**, *12*, 814.
[17] M. Remskar, Z. Skraba, R. Sanjines, F. Levy, *Surf. Rev. Lett.* **1999**, *6*, 1283.
[18] P. M. Voyles, D. A. Muller, J. L. Grazul, P. H. Citrin, H. J. L. Gossmann, *Nature*, **2002**, *416*, 826.
[19] D. O. Dumcenco, H. Kobayashi, Z. Liu, Y. S. Huang, K. Suenaga, *Nat. Comm.* **2013**, *4*, 1351.
[20] W. Zhou, M. D. Kapetanakis, M. P. Prange, S. T. Pantelides, S. J. Pennycook, J. C. Idrobo, *Phys. Rev. Lett.* **2012**, *109*, 206803.
[21] Q. M. Ramasse, C. R. Seabourne, D. M. Kepaptsoglou, R. Zan, U. Bangert, A. J. Scott, *Nano Lett.* **2013**, *13*, 4989.
[22] U. Bangert, W. Pierce, D. M. Kepaptsoglou, Q. Ramasse, R. Zan, M. H. Gass, J. A. Van den Berg, C. B. Boothroyd, J. Amani, H. Hofsäss, *Nano Lett.* **2013**, *13*, 4902.
[23] A. W. Robertson, B. Montanari, K. He, J. Kim, C. S. Allen, Y. A. Wu, J. Olivier, J. Neethling, N. Harrison, A. I. Kirkland, J. H. Warner, *Nano Lett.* **2013**, *13*, 1468.
[24] J. Lee, W. Zhou, S. J. Pennycook, J. C. Idrobo, S. T. Pantelides, *Nat. Commun.* **2013**, *4*, 1650.
[25] J. V. Lauritsen, J. Kibsgaard, G. H. Olesen, P. G. Moses, B. Hinnemann, S. Helveg, J. K. Nørskov, B. S. Clausen, H. Topsøe, E. Lægsgaard, F. Besenbacher, *J. Catal.* **2007**, *249*, 220.
[26] J. Kibsgaard, A. Tuxen, K. G. Knudsen, M. Brorson, H. Topsøe, E. Lægsgaard, J. V. Lauritsen, F. Besenbacher, *J. Catal.* **2010**, *272*, 195.
[27] D. O. Dumcenco, H. P. Hsu, Y. S. Huang, C. H. Liang, K. K. Tiong, C. H. Du, *Mater. Chem. Phys.* **2008**, *111*, 475.
[28] G. Kresse, J. Hafner, *Phys. Rev. B* **1993**, *47*, 558.
[29] G. Kresse, J. Furthmuller, *Comput. Mater. Sci.* **1996**, *6*, 15.
[30] F. Bruneval, *Phys. Rev. Lett.* **2012**, *108*, 256403.
[31] J. Lin, H. Li, H. Zhang, W. Chen, *Appl. Phys. Lett.* **2013**, *102*, 203109.
[32] N. R. Wilson, P. A. Pandey, R. Beanland, R. J. Young, I. A. Kinloch, L. Gong, Z. Liu, K. Suenaga, J. P. Rourke, S. J. York, J. Sloan, *ACS Nano* **2009**, *3*, 2547.
[33] H. P. Komsa, J. Kotakoski, S. Kurasch, O. Lehtinen, U. Kaiser, A. V. Krasheninnikov, *Phys. Rev. Lett.* **2012**, *109*, 035503.
[34] R. Zan, Q. M. Ramasse, R. Jalil, T. Georgiou, U. Bangert, K. S. Novoselov, *ACS Nano* **2013**, *7*, 10167.
[35] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, *77*, 3865.
[36] G. Henkelman, B. P. Uberuaga, H. Jonsson, *J. Chem. Phys.* **2000**, *113*, 9901.

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